F1 Race Intelligence Dashboard

Tableau Project Documentation

An interactive Tableau project for exploring Formula 1 race results, constructor performance, tyre-compound outcomes, driver podiums, and race reliability using the supplied dataset.

Project Overview

This document summarizes the Tableau workbook, dashboard views, data mapping, known limitations, and setup instructions for a GitHub project repository.

Dataset Summary

Source worksheet	Records
race_results	6,464 rows
championship_standings	311 rows
lap_intelligence	3,814 rows
qualifying_results	3,130 rows
circuits	29 rows

The race_results tab contains 6,464 rows and covers seasons 2010–2026 (17 seasons). The mapped race data identifies 334 unique race events.

Dashboard Contents

Dashboard 1 — F1 Race Intelligence

- Constructor Points Trend: points by season and team.
- Tyre Compound Outcomes: points by starting tyre compound, segmented by circuit type.
- DNF Reasons by Team: retirements grouped by team and reason.
- Podiums by Driver: podium finishes by driver.

Dashboard 2 — Broadcast & Reliability

- Driver Points Leaderboard: total points by driver.
- Grand Prix Hosts by Country: distinct race-event count by country.
- Driver Reliability by Driver: average DNF flag as a row-level reliability proxy.
- Podiums by Driver: podium finishes by driver.

Data Mapping

Dashboard field	Source / rule
Race ID	race_id
Grand Prix / Circuit	circuit_name (venue/circuit name supplied in source)
Circuit Country	country
Circuit Type	circuit_type
Tyre Compound	starting_compound
Podium Flag	1 when finish_position is 1, 2, or 3
Winner Flag	1 when finish_position is 1
DNF Flag	dnf
Status	DNF when dnf = 1; otherwise Finished

Known Limitations

- Driver nationality and team nationality are not supplied in the uploaded workbook; nationality fields are left blank and are not used in the dashboards.
- The mapped Grand Prix label uses circuit_name because the source does not provide a separate official event-name field.
- Driver reliability is shown as an average of the DNF flag, so interpret it as a row-level DNF-rate proxy.
- The workbook XML and packaged ZIP structure were checked, but interactive rendering should still be confirmed in Tableau Desktop.

Suggested Calculated Fields

Field	Formula
DNF Rate	SUM([DNF Flag]) / COUNT([Race ID])
Podium Count	SUM([Podium Flag])
Points per Start	SUM([Points]) / COUNT([Race ID])
Positions Gained	AVG([Positions Gained]) — where this field exists in the connected data

Open the Project in Tableau

1. Download or clone the repository, including the packaged .twbx workbook and supporting files.
2. Open Tableau Desktop (or a compatible Tableau application).
3. Choose File > Open and select the packaged Tableau workbook (.twbx).
4. If prompted, allow the workbook to upgrade and check that the packaged CSV data source is available.
5. Review worksheets and dashboards, then save a local copy after confirming the views render correctly.

Suggested GitHub Repository Contents

- F1_Race_Intelligence_UserData_Recolored.twbx — packaged Tableau workbook
- F1_Race_Intelligence_Project.pptx — project presentation
- F1_Race_Intelligence_Project.pdf — shareable project brief
- F1_Race_Intelligence_Project.docx — editable project documentation
- README.md — repository landing page

Distribution note: Only include source data files in a public repository if you have permission to redistribute them.